

Course Title: Microcomputer Application I
Course Code: CSE-2402

Topic: Different topics of windows

(Files, Folders, Basic operation on files and folders, searching file and folders)

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Content of Mid-term Exam

- 1) Different topics of windows: Files, Folders, Basic operations on file/folders, Searching file and folders**
- 2) Microsoft Word topics: Formatting, Table Editing**
- 3) Microsoft Word topics: Picture, Clip-art and object, Drawing, Text box and shapes**
- 4) Microsoft Word topics: Hyperlink, Macro**
- 5) Discussion on Microsoft Word related assignment**

Learning Objectives

After this lecture, students will be able to:

- Understand Microcomputer and Microprocessor
- Explain Files and Folders
- Learn basic operations on files and folders

Introduction to Microcomputer

- ✓ A microcomputer is a small and affordable computer made for personal use. Some people also call it a micro computer. Its main part is a microprocessor. It includes desktops, laptops, and tablets. People use them for studying, playing games, and doing work. Micro personal computers are widely used in homes, schools, and offices



Types of Microcomputers

Microcomputers are small and low-cost computers used for school, home, and office work. Here are the main types of microcomputers:

1. Desktops: People use these for work or study at one place.
2. Laptops: Light devices used for watching videos or browsing
3. Tablets: People watch videos or browse the internet on lightweight tablets.
4. Workstations: These are powerful computers for big tasks like design.
5. Gaming Consoles: Gamers play on devices like PlayStation or Xbox.

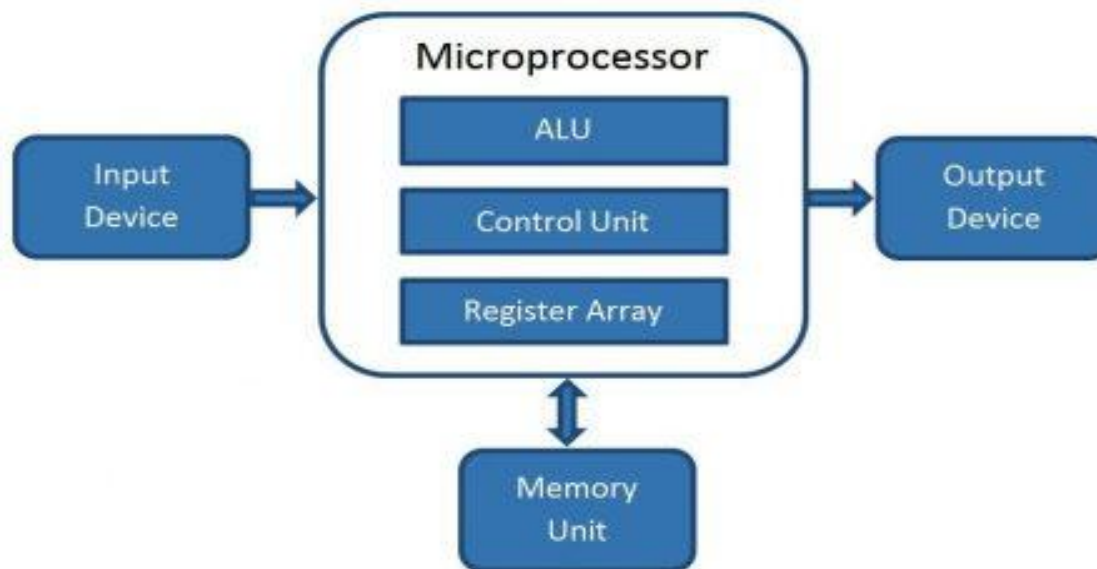
Parts of a Microcomputer

- CPU: It controls the computer.
- RAM: It helps the computer remember things.
- Storage: It saves your files.
- Input Devices: You use them to give commands (keyboard, mouse).
- Output Devices: They show the results (monitor, speakers).

Introduction to Microprocessor

- ✓ A microprocessor is an electronic integrated circuit (IC) that acts as the central processing unit (CPU) of a computer. It is responsible for executing instructions, performing arithmetic and logical operations, and controlling the overall functioning of a computer system.
- ✓ Key Functions of a Microprocessor
 - 1) Arithmetic operations: Addition, subtraction, multiplication, division
 - 2) Logical operations: AND, OR, NOT, comparisons
 - 3) Control operations: Directs data flow between memory, input, and output devices
 - 4) Instruction execution: Fetches, decodes, and executes instructions

Components of Microprocessor



Components of Microprocessor

Input Device:

This is where data or instructions are entered into the computer.

Examples: keyboard, mouse.

Microprocessor (CPU):

This is the brain of the computer. It has three main parts:

ALU (Arithmetic Logic Unit): Performs calculations like addition, subtraction, and comparisons.

Control Unit: Controls and coordinates all operations. It tells other parts what to do.

Register Array: Small, fast memory inside the CPU used to store temporary data.

Memory Unit:

Stores data and instructions needed for processing. The microprocessor reads from and writes to memory.

Output Device:

Shows the result after processing.

Examples: monitor, printer.

Applications of Microcomputers

Microcomputers are widely used in offices, educational institutions, and organizations through application software. Among the most common and important applications are MS Word, MS Excel, and MS PowerPoint, which are part of the Microsoft Office suite.

Microcomputers are used in many fields because of their versatility and efficiency. Some common applications are:

- ☐ **Education:** Online learning, presentations, assignments, and research
- ☐ **Business:** Accounting, payroll, inventory management, and office automation
- ☐ **Communication:** Email, video conferencing, social media
- ☐ **Healthcare:** Patient record management, diagnostic systems
- ☐ **Entertainment:** Games, music, movies, and multimedia
- ☐ **Scientific and Engineering Work:** Data analysis, simulations, and design

What is Windows?

- Windows is an Operating System developed by Microsoft.
(An Operating System(OS) is a software that manages and handles hardware and software resources of a computing device.)
- It acts as an interface between user and computer hardware.
- Provides a Graphical User Interface (GUI).

Examples:

Windows 10

Windows 11

Why We Use Windows

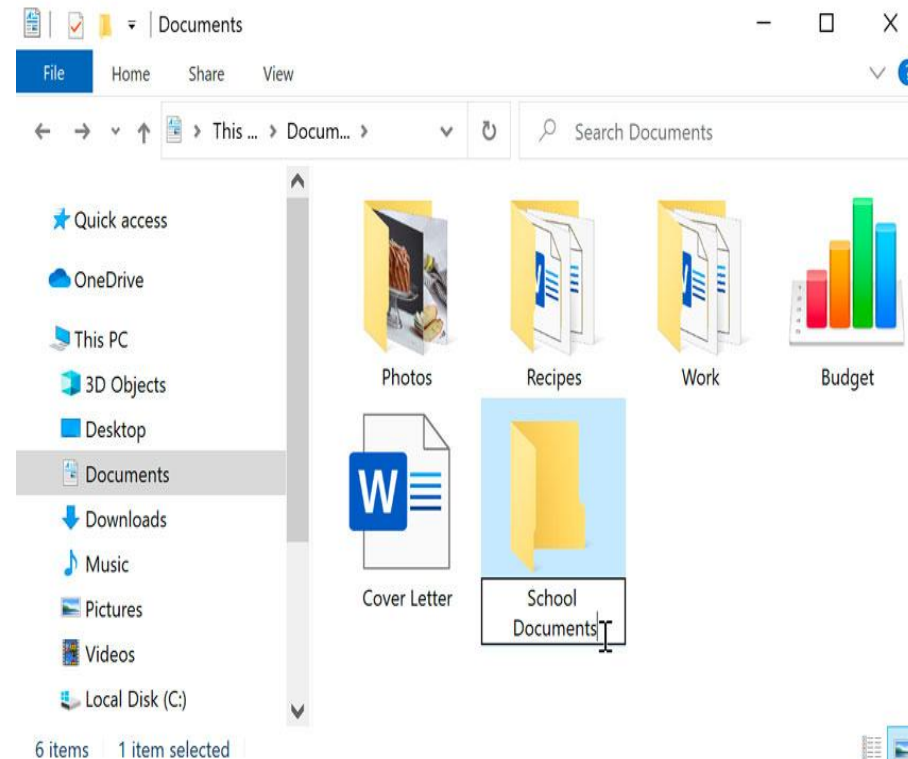
- Easy to use graphical environment
- Supports multitasking
- Manages files, folders, memory, and devices
- Widely used in offices, universities, and homes

Main Parts of Windows Interface

- Desktop – Main working area
- Icons – Represent programs, files, folders
- Taskbar – Shows running applications
- Start Menu – Access programs and system settings

What is a File?

- Files can be described as a set of related data/information and is stored on secondary storage device.
- Every file has: File name and File extension
- Examples:
 - assignment.docx → Word file
 - photo.jpg → Image file
 - song.mp3 → Audio file

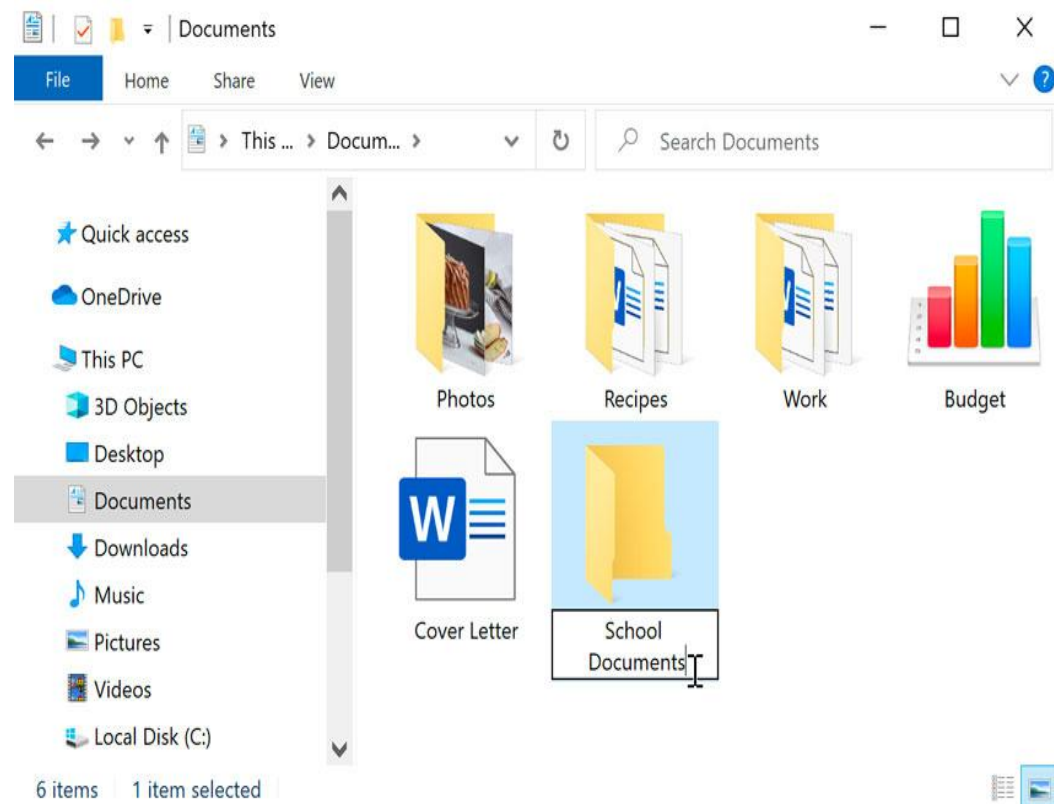


Types of Files

- Document files: .docx, .pdf, .txt
- Image files: .jpg, .png
- Audio files: .mp3, .wav
- Video files: .mp4, .avi
- Program files: .exe

What is a Folder?

- A Folders can be described as a container to store files. Thus, in simple words, a folder is a collection of related files stored in the computer memory.
- A folder can contain:
 - Files
 - Subfolders
- Example:
 - Documents
 - Semester 1
 - ICT
 - Notes.docx



Importance of Folders

- Keeps files organized
- Saves time
- Easy searching
- Prevents confusion and data loss

Difference between File and Folder

Factor	File	Folder
Extension	Files may or may not have extensions.	Folders do not have extensions.
Container	A file cannot contain another file/folder.	A folder can contain any number of file/folders.
Memory size	A file has certain size and memory consumption.	A folder has no size of its own. It derives the size from the files it contains.
Attributes	Name, Extension, Date, Time, Length and Protection (Read-Only, hidden etc.)	Name, Date, Time and Protection (Read-Only, hidden etc.)
Supported operations	Open, save, rename, print, email and modify file content.	Move, rename, delete and share.

Practice Questions

- Q: Can a file contain another file?
- A: No. Only folders can contain files. A file stores data, not other files.

- Q: Is a folder useless if it is empty?
- A: No. An empty folder is still useful for future organization.

- Q: Is a folder also a file?
- A: No. A folder is a container, while a file stores data.